# SMRTTECH 4AI3
# Course Project

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# Housekeeping

- Students to form teams of 2-4
- Define a real-world problem, study it together and solve it with ML

# Deliverables

- Group presentation
- Recorded video clip for project demo
- Source code (Jupiter Notebook is preferred)
- Written report (PDF, shorter is better as long as the explanation is clear, try to avoid using large tables, images, bullet points.)

# Project Assessment

- Projects will be assessed based on students' understanding and model performance.
- Each member on a team will be assessed individually.
  - Please include a list of responsibility and contribution for each team member, and attach it in the beginning of your presentation.

For example:

- Name: John Doe
- Responsible: Model Training
- Contribution: Trained and evaluated the classification model, tuned hyperparameters, and reported the final test results.

# Some Ideas and an Appetizer for you

Tier 1) - Less Challenging

Find an appropriate dataset online. Train your own models. Report on results and differences.

- The UCI ML Repo: https://archive.ics.uci.edu/
- Kaggle: https://www.kaggle.com/datasets

Tier 2) - More Challenging

Build your own dataset. Finetune a pretrained model to solve the task.

- Hugging Face: https://huggingface.co/models (Students can utlilize their Trainer API for training)
- Ultralytics: https://github.com/ultralytics/ultralytics (This is where to find the YOLO models)
- Roboflow: https://roboflow.com/ (For building Computer Vision datasets)

Tier 3) - Most Challenging

Delve deeper into studying different models' architectures. Select a dataset and compare two or more model (with different architectures) and investigate how their architectural priors affect task performance. (Eg. Compare MLP with CNN for Image Recognition to study the effect of the Translation-Invariance prior in a CNN). Or select a specific architectural trick or component from the literature, understand how and why it works, and experimentally demonstrate its effect by comparing model performance with and without this component (Eg. Comparing neural networks with and without attention).
https://github.com/saberxcyu/Transformer-LSTM-NMT

Any questions?